

Advanced Structural Analysis using ANSYS

Lecture and Exercise Notes

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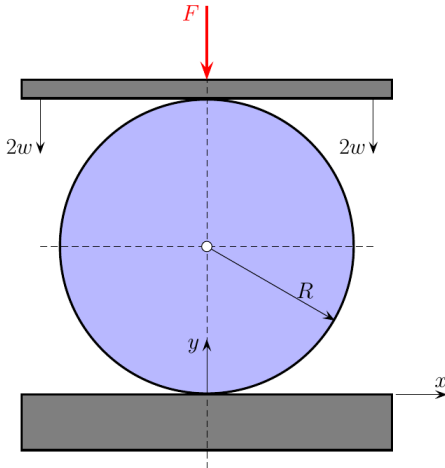
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Exercise 4: Contact Simulation

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

HERTZ Contact Problem

Problem description



- Geometric data
 $R = 100 \text{ mm}$ (Radius)
- Material data of the ball (steel)
 $E = 210,000 \text{ N/mm}^2$
 $\nu = 0.3$
 contact surfaces are rigid
- Load
 $2w = 0.2 \text{ mm}$
- Objective target
 Reaction force F ,
 Contact radius a ,
 Maximum area stress p_0

Analytical Description of Problem

HERTZ Contact, Problem 1

A linear-elastic ball in frictionless contact with a non-deformable half space

Reference: H. HERTZ: *Über die Berührung fester elastischer Körper. Journal für die reine und angewandte Mathematik*, (1882).

Deformation of the ball in the contact zone

$$w = \sqrt[3]{\left(\frac{3F(1-\nu^2)}{4E}\right)^2 \frac{1}{R}}$$

Reaction force

$$F = \frac{4E\sqrt{R}}{3(1-\nu^2)} w^{\frac{3}{2}}$$

Radius of the contact area

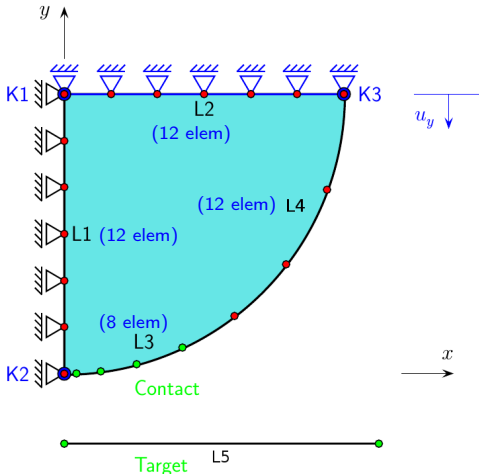
$$a = \sqrt[3]{\frac{3F(1-\nu^2)R}{4E}}$$

Maximum pressure

$$p_0 = \frac{1}{\pi} \sqrt[3]{\frac{6FE^2}{R^2(1-\nu^2)^2}}$$

3D model, boundary condition, load

HERTZ Contact, Problem 1



• Element types

Ball segment:

- 2-D PLANE182
- (Simplified enhanced strain formulation)
- (axially symmetric)

Ground:

- automatic selection of contact element

• Boundary conditions

Nodes	Condition
$x = 0$	$u_x = 0$
$y = R$	$u_y = -1.1 \text{ mm}$

• Meshing

- Division of arc 3 : 7 into Line L3 and L4
- Freely meshed
- Maximum mesh refinement at keypoint 2
- Contact modeling using the Contact Wizard
- Generation of node components
- Initial distance 1 mm

• Solution

1. LS: closing contact; 2. LS: deformation

Additional ANSYS APDL Commands

HERTZ Contact, Problem 1

Pre Processing

/PREP7

(geometrical) division of lines **LDIV**,...

local mesh refinement **KREFINE**,...

Post Processing

/POST1

print reaction forces (only bc's) **PRRFOR**,...

print reaction forces (all) **PRRSOL**,...

General Commands

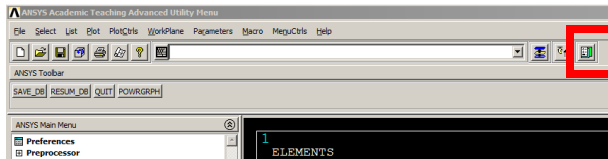
define component **CM**,...

select component **CMSEL**,...

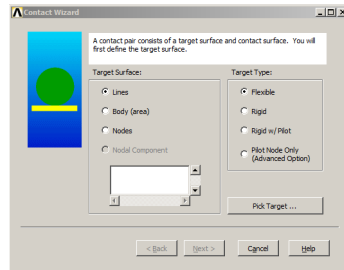
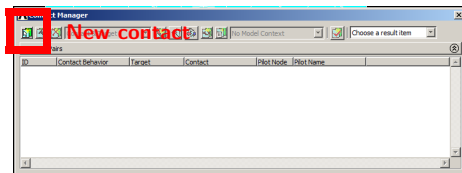
more help: help,XXX, whereby XXX should be replace by command.

Contact definition using *Contact Wizard*

HERTZ Contact, Problem 1

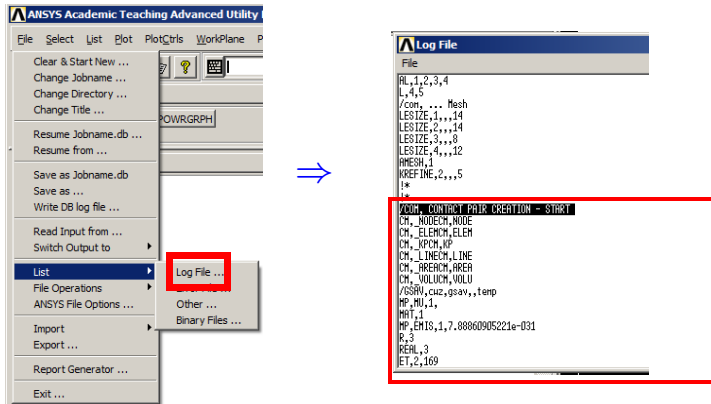


Contact wizard



Contact definition using *Contact Wizard* (contd.)

HERTZ Contact, Problem 1



Copy output between

/COM, CONTACT PAIR CREATION - START and */COM, CONTACT PAIR CREATION - END*
from logfile jobfile.out and paste it to your APDL-Script.

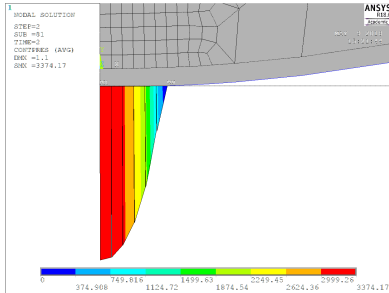
Important: This script part has static numbering and parameters.

⇒ Any change to the part before the contact definition, which cause e.g. a renumbering of nodes/elements, requires a rebuild of the contact definition.

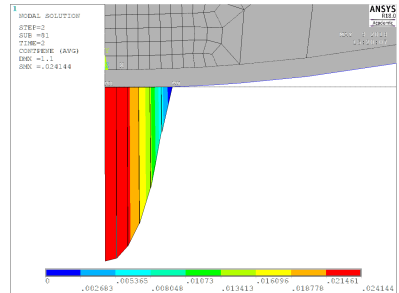
Checking Results

HERTZ Contact, Problem 1

contact pressure



contact penetration



		ANSYS	analytical
max. contact pressure	p_0 in N/mm ²	3374	4502

- Bad representation of the contact situation
- max. penetration of 0.024 mm compared to maximal possible penetration of 0.1 mm (wrt. to no-acting contact) is 25 %

⇒ Modification of Contact Stiffness Parameter

Results with modified contact stiffness

HERTZ Contact, Problem 1

Insert

```
RMODIF,3,3,10
```

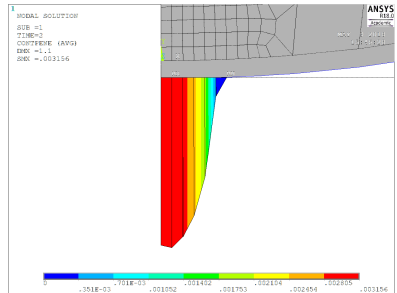
after

```
/COM, CONTACT PAIR CREATION - END
```

It sets the contact stiffness parameter to 10 by modification of the Real Constant data of the contact target.

This results in a max. penetration of 0.0032 mm, which is 0.3 % compared to maximal possible penetration of 0.1 mm (wrt. to no-acting contact)

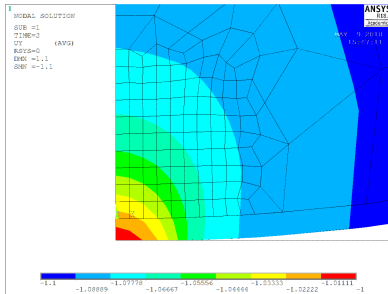
contact penetration



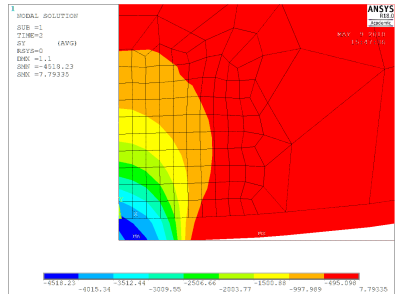
Results with modified contact stiffness (contd.)

HERTZ Contact, Problem 1

displacement y -direction



stress y -direction

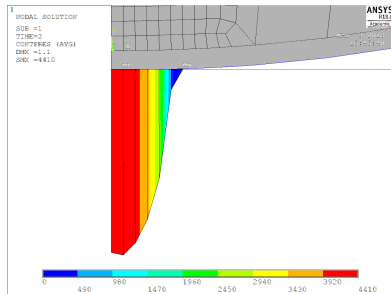


		ANSYS	analytical
reaction force	F in N	96662	97301
contact radius	a in mm	3.9262	3.16
max. contact pressure	p_0 in N/mm ²	4415	4502

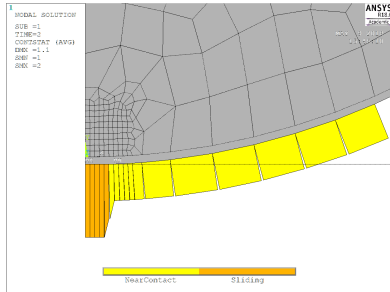
Results with modified contact stiffness (contd.)

HERTZ Contact, Problem 1

contact pressure



contact state



		ANSYS	analytical
reaction force	F in N	96662	97301
contact radius	a in mm	3.9262	3.16
max. contact pressure	p_0 in N/mm²	4415	4502

Additional tasks

HERTZ Contact, Problem 1

Try on yourself the following tasks and see what are the change in results

- Modify the displacement of the ball (“more contact”)
- Create a animation/movie of the simulation
- Change the contact properties (stiffness, max. penetration, ...)
- “own idea” ???